Numere complexe. Funcții complexe.

(1) Determinați forma trigonometrică / exponențială $z = r(\cos t + i\sin t) = re^{it}$ a urm. nr. complexe și reprezentați în plan punctele ale căror afixe sunt aceste nr:

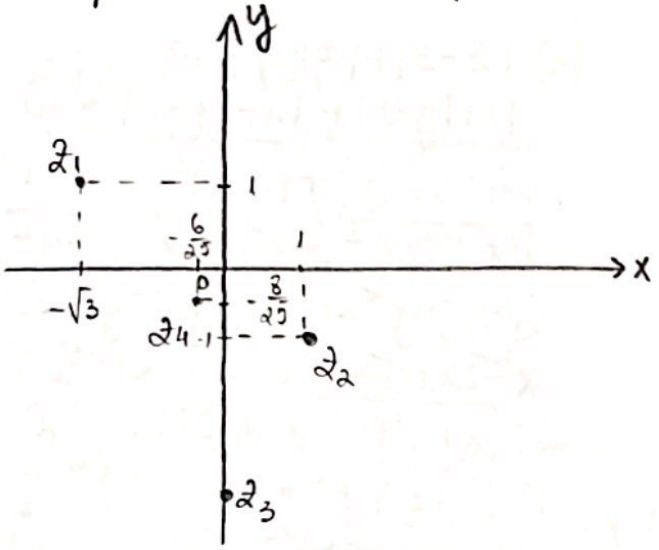
(a) $z_1 = -\sqrt{3}+i$ (b) $z_2 = 1-i$ (c) $z_3 = -3i$

(d) $z_4 = \frac{2}{-3+4i}$

Rezolvare

$$z_4 = \frac{2}{-3+4i} = \frac{2(3+4i)}{(4i+3)(4i-3)} = \frac{6+8i}{-25} = -\frac{6}{25} - \frac{8}{25}i.$$

Reprezentăm în plan cele 4 puncte:



- pt z_1: $|z_1| = \sqrt{3+1} = 2$

$\text{Arg}(z_1) = \pi + \text{arctg}\frac{1}{-\sqrt{3}} = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

$\Rightarrow z_1 = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right) = 2e^{\frac{5i\pi}{6}}$

- pt z_2: $|z_2| = \sqrt{1+1} = \sqrt{2}$

$\text{Arg}(z_2) = \text{arctg}\frac{1}{-1} = -\frac{\pi}{4}$

$\Rightarrow z_2 = \sqrt{2}\left(\cos\frac{-\pi}{4} + i\sin\frac{-\pi}{4}\right) = \sqrt{2}e^{\frac{7i\pi}{4}}$

- pt z_3: $|z_3| = \sqrt{0+9} = 3$

$\text{Arg}(z_3) = -\frac{\pi}{2}$

$\Rightarrow z_3 = 3\left(\cos\frac{-\pi}{2} + i\sin\frac{-\pi}{2}\right) = 3e^{\frac{3i\pi}{2}}$

- pt z_4: $|z_4| = \sqrt{\frac{36}{625} + \frac{64}{625}} = \frac{10}{25} = \frac{2}{5}$

$\text{Arg}(z_4) = \pi + \text{arctg}\frac{4}{3}$

$\Rightarrow z_4 = \frac{2}{5}\left(\cos(\text{arctg}\frac{4}{3} + \pi) + i\sin(-11-)\right) = \frac{2}{5}e^{i(\text{arctg}\frac{4}{3}+\pi)}$

(2.) Reprezentați grafic mulțimile din planul complex care verifică fiecare dintre relațiile de mai jos:

(a) $|z+1-i| = 1$

(b) $|z-1| = |z-i|$

(c) $|z+\sqrt{2}i| < 2$

(d) $1 \le |z+1+\sqrt{3}i| \le 2$

(e) $|z+3| + |z-3| = 10$

(f) $-1 \le \text{Im}(z) \le 3$

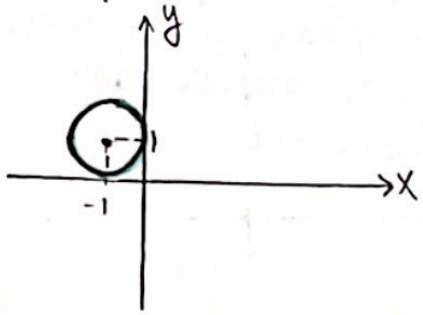
(g) $\text{Re}(z) \neq 0$

(h) $2|z-i| = |z|$.

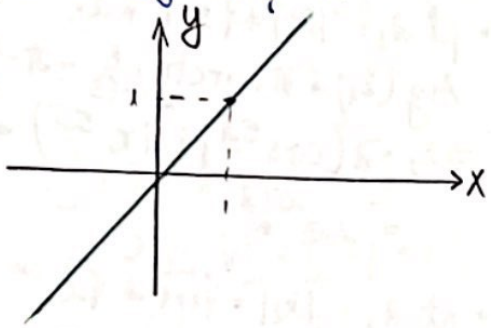
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Rezolvare

(a) $|z+1-i|=1$ este cercul cu centrul în pct de afix $-1+i$ și de rază 1.



(b) $|z-1|=|z-i|$ este dreapta mediatoare a segm ce unește pct de afix 1 și i.



$|z-1|=|z-i| \Leftrightarrow |x+iy-1| = |x+iy-i| \Leftrightarrow$

$\Leftrightarrow \sqrt{(x-1)^2+y^2} = \sqrt{x^2+(y-1)^2} \uparrow^2$

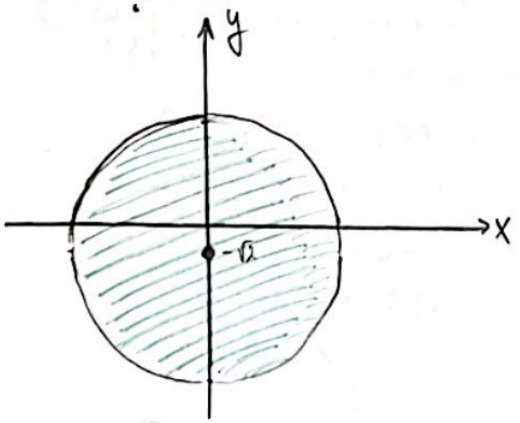
$(x-1)^2+y^2 = x^2+(y-1)^2$

$\cancel{x^2}-2x+1+\cancel{y^2} = \cancel{x^2}+\cancel{y^2}-2y+1$

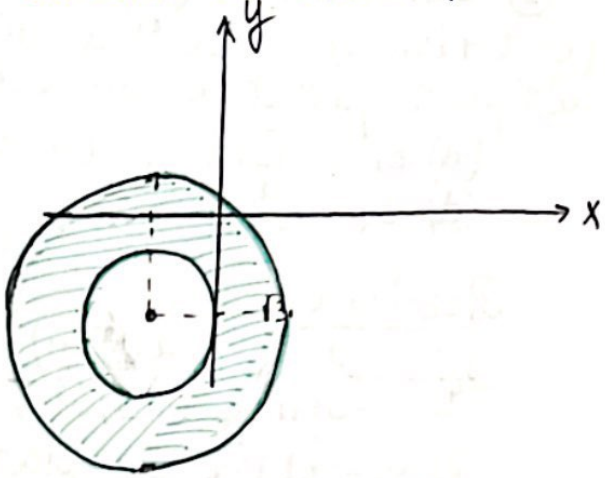
$-2x = -2y \mid :(-2)$

$x=y$ (prima bisectoare)

(c) $|z+\sqrt{2}i|<2$ este disc cu centrul în pct de afix $-\sqrt{2}i$ și de rază 2.



(d) $1 \le |z+1+\sqrt{3}i| \le 2$ este coroană circulară centrată în pct de afix $-1-\sqrt{3}i$, cuprinsă între discurile de rază 1 și 2.



(e) $|z-3|+|z+3|=10$

$|x+iy-3|+|x+iy+3|=10$

$\sqrt{(x-3)^2+y^2}+\sqrt{(x+3)^2+y^2}=10$

$\sqrt{(x-3)^2+y^2} = 10-\sqrt{(x+3)^2+y^2} \uparrow^2$

$\cancel{(x-3)^2}+\cancel{y^2} = 100+\cancel{(x+3)^2}+\cancel{y^2}-20\sqrt{(x+3)^2+y^2}$

$\cancel{x^2}-6x+\cancel{9} = 100+\cancel{x^2}+6x+\cancel{9}-20\sqrt{(x+3)^2+y^2}$

$20\sqrt{(x+3)^2+y^2} = 100+12x \mid :2$

$10\sqrt{(x+3)^2+y^2} = 50+6x \uparrow^2$

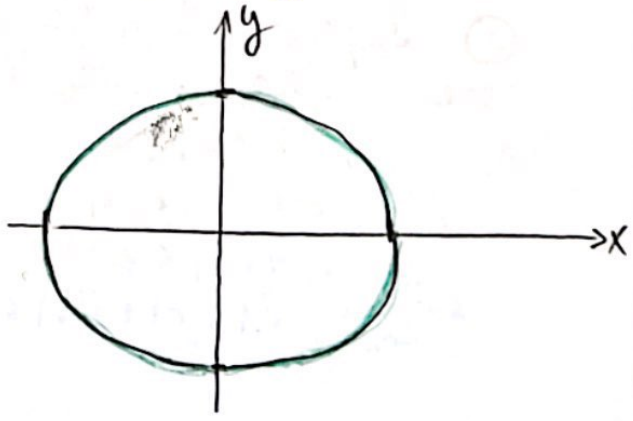
$100[(x+3)^2+y^2] = 2500+36x^2+600x$

$100x^2+600x+900+100y^2 = 2500+36x^2+600x$

$64x^2+100y^2 = 1600 \mid :4$

$16x^2+25y^2 = 400 \mid :400$

$\frac{x^2}{25}+\frac{y^2}{16} = 1$ este elipsa cu axele 5 și 4 și focarele 3 și -3.

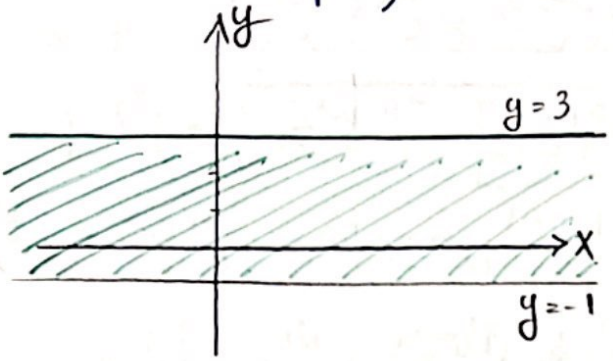


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(f) $-1 < \mathrm{Im}(z) \leq 3$

$\mathrm{Im}(z) = \mathrm{Im}(x+iy) = y$

$\Rightarrow -1 < y \leq 3$ este banda orizontală determinată de dreptele $y=-1$ și $y=3$ (inclusiv a doua dreaptă).



(g) $\mathrm{Re}(z) \neq 0$ este planul complex, mai puțin axa Oy.



(h) $2|z-i| = |z|$

$2|x+iy-i| = |x+iy|$

$2\sqrt{x^2+(y-1)^2} = \sqrt{x^2+y^2}$ $\uparrow^2$

$4x^2+4y^2-8y+4 = x^2+y^2$

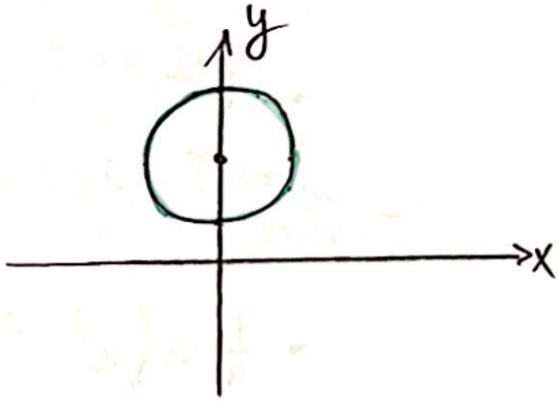
$3x^2+3y^2-8y+4 = 0 \mid :3$

$x^2+y^2-\frac{8}{3}y+\frac{4}{3} = 0$

$x^2+y^2-\frac{8}{3}y+\frac{16}{9}-\frac{16}{9}+\frac{4}{3} = 0$

$x^2+\left(y-\frac{4}{3}\right)^2 = \frac{16}{9}-\frac{4}{3} = \frac{4}{9} = \left(\frac{2}{3}\right)^2$

este ecuația cercului cu centrul în pct de afix $\frac{4}{3}i$ și raza $\frac{2}{3}$



3. Pornind de la definiția funcțiilor elementare, verificați că formulele cunoscute

$$\cos(z \pm w) = \cos z \cos w \mp \sin z \sin w$$
$$\sin(z \pm w) = \sin z \cos w \pm \cos z \sin w$$

rămân valabile pentru $\forall z, w \in \mathbb{C}$.

Funcții elementare: $e^z = e^x(\cos y + i \sin y)$;

$$\cos z = \frac{e^{iz}+e^{-iz}}{2}; \quad \sin z = \frac{e^{iz}-e^{-iz}}{2i}$$

$$\mathrm{ch}\, z = \frac{e^z+e^{-z}}{2}; \quad \mathrm{sh}\, z = \frac{e^z-e^{-z}}{2}.$$

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Rezolvare

$\cos(z+w) = \cos z \cos w - \sin z \sin w =$

$= \frac{e^{iz}+e^{-iz}}{2} \cdot \frac{e^{iw}+e^{-iw}}{2} - \frac{e^{iz}-e^{-iz}}{2i} \cdot \frac{e^{iw}-e^{-iw}}{2i} =$

$= \frac{e^{i(z+w)} + e^{i(w-z)} + e^{i(z-w)} + e^{-i(z+w)}}{4} +$

$+ \frac{e^{i(z+w)} - e^{i(w-z)} - e^{i(z-w)} + e^{-i(z+w)}}{4} =$

$= \frac{1}{4} \cdot 2 \left[e^{i(z+w)} + e^{-i(z+w)} \right] =$

$= \frac{1}{2} \left(e^{i(z+w)} + e^{-i(z+w)} \right) = \cos(z+w)$ (Adevărat)

$\cos(z-w) = \cos z \cos w + \sin z \sin w =$

$= \frac{e^{iz}+e^{-iz}}{2} \cdot \frac{e^{iw}+e^{-iw}}{2} + \frac{e^{iz}-e^{-iz}}{2i} \cdot \frac{e^{iw}-e^{-iw}}{2i} =$

$= \frac{e^{i(z+w)} + e^{i(w-z)} + e^{i(z-w)} + e^{-i(z+w)}}{4} -$

$- \frac{e^{i(z+w)} - e^{i(w-z)} - e^{i(z-w)} + e^{-i(z+w)}}{4} =$

$= \frac{1}{4} \cdot 2 \left(e^{i(w-z)} + e^{i(z-w)} \right) =$

$= \frac{1}{2} \left(e^{i(z-w)} + e^{-i(z-w)} \right) = \cos(z-w)$ (Adevărat).

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(4.) Determinați partea reală, partea imaginară și modulul pentru fiecare din următoarele funcții complexe.

(a) e^{2z} (b) e^{z^2} (c) $\cos z$ (d) e^{iz}

(e) $2e^{1-z}$ (f) $\sin z$ (g) $\operatorname{ch} z$ (h) $\operatorname{sh} z$.

Rezolvare

(a) $e^{2z} = e^{2x+2iy} = e^{2x}(\cos 2y + i \sin 2y) = e^{2x}\cos 2y + i \sin 2y \cdot e^{2x}$.

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$\Rightarrow Re(e^{2z}) = e^{2x}\cos 2y \; ; \; Im(e^{2z}) = e^{2x}\sin 2y$

$|e^{2z}| = \sqrt{e^{4x}\cos^2 2y + e^{4x}\sin^2 2y} = \sqrt{e^{4x}} = e^{2x}$

(b) $e^{z^2} = e^{(x+iy)^2} = e^{x^2-y^2+2ixy} = e^{x^2-y^2}(\cos 2xy + i\sin 2xy) =$

$= e^{x^2-y^2}\cos 2xy + i e^{x^2-y^2}\sin 2xy$

$\Rightarrow Re(e^{z^2}) = e^{x^2-y^2}\cos 2xy \; ; \; Im(e^{z^2}) = e^{x^2-y^2}\sin 2xy$

$|e^{z^2}| = \sqrt{e^{[x^2-y^2]^2}\cos^2 2xy + e^{[x^2-y^2]^2}\sin^2 2xy} = e^{x^2-y^2}$

(c) $\cos z = \cos(x+iy) = \left[e^{i(x+iy)} + e^{-i(x+iy)}\right]\cdot\frac{1}{2} =$

$= \frac{1}{2}\left[e^{-y+ix} + e^{y-ix}\right] = \frac{1}{2}\left[e^{-y}(\cos x + i\sin x) + e^{y}(\cos(-x) + i\sin(-x))\right]$

$= \frac{1}{2}\left[e^{-y}(\cos x + i\sin x) + e^{y}(\cos x - i\sin x)\right] =$

$= \frac{1}{2}\left[\cos x\,(e^{-y} + e^{y}) + i\sin x\,(e^{-y} - e^{y})\right] =$

$= \cos x \operatorname{ch} y - i\sin x \operatorname{sh} y$

$\Rightarrow Re(\cos z) = \cos x \operatorname{ch} y \; ; \; Im(\cos z) = -\sin x \operatorname{sh} y$

$|\cos z| = \sqrt{\cos^2 x \operatorname{ch}^2 y + \sin^2 x \operatorname{sh}^2 y}$

(d) $e^{iz} = e^{i(x+iy)} = e^{-y+ix} = e^{-y}(\cos x + i\sin x) =$

$= e^{-y}\cos x + i e^{-y}\sin x$

$\Rightarrow Re(e^{iz}) = e^{-y}\cos x \; ; \; Im(e^{iz}) = e^{-y}\sin x$

$|e^{iz}| = \sqrt{e^{-2y}\cos^2 x + e^{-2y}\sin^2 x} = e^{-y}$

(e) $\sin z = \sin(x+iy) = \frac{1}{2i}\left[e^{i(x+iy)} - e^{-i(x+iy)}\right] =$

$= \frac{1}{2i}\left(e^{-y+ix} - e^{y-ix}\right) = \frac{1}{2i}\left[e^{-y}(\cos x + i\sin x) - e^{y}(\cos(-x) +\right.$

$\left. + i\sin(-x))\right] = \frac{1}{2i}\left[e^{-y}(\cos x + i\sin x) - e^{y}(\cos x - i\sin x)\right] =$

$= \frac{1}{2i}\left[\cos x\,(e^{-y} - e^{y}) + i\sin x\,(e^{-y} + e^{y})\right] =$

$= i\sin x \operatorname{ch} y + \cos x \operatorname{sh} y.$

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$\Rightarrow \operatorname{Re}(\sin z) = \sin x \operatorname{ch} y \; ; \; \operatorname{Im}(\sin z) = \cos x \operatorname{sh} y$

$|\sin z| = \sqrt{\sin^2 x \operatorname{ch}^2 y + \cos^2 x \operatorname{sh}^2 y}$

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(5.) Determinați soluțiile următoarelor ecuații în mulțimea numerelor complexe.

(a) $e^z = -1$ (b) $e^z = 2i$ (c) $\sin z = \frac{5}{3}$

(d) $\operatorname{sh} z = \frac{i}{2}$ (e) $\cos z = \frac{3+i}{4}$ (f) $\operatorname{ch} z = \frac{1}{2}$

(g) $z^2 + 2i = 0$ (h) $z^6 - 64 = 0$ (i) $z^4 + 7 - 24i = 0$.

Rezolvare

(a) $e^z = -1 \Leftrightarrow z = \operatorname{Ln}(-1) = \{\underbrace{\ln 1}_{0} + i[\operatorname{Arg}(-1) + 2k\pi] \mid k \in \mathbb{Z}\} \Leftrightarrow$

$\Leftrightarrow z = \{i(\pi + 2k\pi) \mid k \in \mathbb{Z}\} \Leftrightarrow z = \{i(2k+1)\pi \mid k \in \mathbb{Z}\}$.

(b) $e^z = 2i \Leftrightarrow z = \operatorname{Ln}(2i) = \{\ln|2i| + i[\operatorname{Arg}(2i) + 2k\pi] \mid k \in \mathbb{Z}\} \Leftrightarrow$

$\Leftrightarrow z = \{\ln 2 + i\left(\frac{\pi}{2} + 2k\pi\right) \mid k \in \mathbb{Z}\} \Leftrightarrow z = \{\ln 2 + i \cdot \frac{1}{2}(4k+1)\pi \mid k \in \mathbb{Z}\}$.

(c) $\sin z = \frac{5}{3} \Leftrightarrow \frac{e^{iz} - e^{-iz}}{2i} = \frac{5}{3} \Leftrightarrow 3e^{iz} - 3e^{-iz} = 10i$

Notăm: $t = e^{iz}$

$\Rightarrow 3t - \frac{3}{t} = 10i \mid \cdot t \Rightarrow 3t^2 - 10it - 3 = 0$

$\Delta = -100 + 36 = -64 \Rightarrow \sqrt{\Delta} = 8i$

$\Rightarrow t_{1,2} = \frac{10i \pm 8i}{6} \rightarrow t_1 = \frac{18i}{6} = 3i$

$t_2 = \frac{2i}{6} = \frac{1}{3}i = 3^{-1}i$

$\Rightarrow e^{iz} = 3^{\pm 1}i \Leftrightarrow iz = \operatorname{Ln}[3^{\pm 1}i] \Leftrightarrow z = -i \operatorname{Ln}[3^{\pm 1}i] \Leftrightarrow$

$\Leftrightarrow z = -i\left[\ln(3^{\pm 1}) + i\left(\frac{\pi}{2} + 2k\pi\right)\right] \; ; k \in \mathbb{Z}$

$z = \frac{\pi}{2} + 2k\pi - i \ln(3^{\pm 1}) \; ; k \in \mathbb{Z}$

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(d) $\operatorname{sh} z = \frac{i}{2} \Leftrightarrow \frac{e^z - e^{-z}}{2} = \frac{i}{2} \Leftrightarrow e^z - e^{-z} = i$

Notăm: $e^z = t$

$\Rightarrow t - \frac{1}{t} = i \mid \cdot t \Rightarrow t^2 - it - 1 = 0$

$\Delta = -1 + 4 = 3 \Rightarrow \sqrt{\Delta} = \sqrt{3}$

$t_{1,2} = \frac{i \pm \sqrt{3}}{2}$

Caz I $t = \frac{i+\sqrt{3}}{2} \Rightarrow e^z = \frac{i}{2} + \frac{\sqrt{3}}{2} \Rightarrow z = \operatorname{Ln}\left(\frac{i}{2} + \frac{\sqrt{3}}{2}\right) \Rightarrow$

$\Rightarrow z = \left\{ \ln \left| \frac{\sqrt{3}}{2} + \frac{1}{2} i \right| + i \left(\operatorname{Arg}\left(\frac{\sqrt{3}}{2} + \frac{1}{2} i\right) + 2k\pi \right) \mid k \in \mathbb{Z} \right\} =$

$= \left\{ \ln 1 + i \left(\operatorname{arctg} \frac{1}{\sqrt{3}} + 2k\pi \right) \mid k \in \mathbb{Z} \right\} =$

$= \left\{ i \left(\frac{\pi}{6} + 2k\pi \right) \mid k \in \mathbb{Z} \right\} = \left\{ i \left(\frac{1}{6} + 2k \right)\pi \mid k \in \mathbb{Z} \right\}$

Caz II $t = \frac{i-\sqrt{3}}{2} \Rightarrow e^z = \frac{i}{2} - \frac{\sqrt{3}}{2} \Rightarrow z = \operatorname{Ln}\left(\frac{i}{2} - \frac{\sqrt{3}}{2}\right) \Rightarrow$

$\Rightarrow z = \left\{ \ln \left| -\frac{\sqrt{3}}{2} + \frac{1}{2} i \right| + i \left[\operatorname{Arg}\left(-\frac{\sqrt{3}}{2} + \frac{1}{2} i\right) + 2k\pi \right] \mid k \in \mathbb{Z} \right\} =$

$= \left\{ \ln 1 + i \left(\pi + \operatorname{arctg} \frac{1}{-\sqrt{3}} + 2k\pi \right) \mid k \in \mathbb{Z} \right\} =$

$= \left\{ i \left(\pi - \frac{\pi}{6} + 2k\pi \right) \mid k \in \mathbb{Z} \right\} =$

$= \left\{ i \left(\frac{5\pi}{6} + 2k\pi \right) \mid k \in \mathbb{Z} \right\} = \left\{ i \left(\frac{5}{6} + 2k \right)\pi \mid k \in \mathbb{Z} \right\}.$

(e) $\cos z = \frac{3+i}{4} \Leftrightarrow \frac{e^{iz} + e^{-iz}}{2} = \frac{3+i}{4} \Leftrightarrow 2e^{iz} + 2e^{-iz} = 3+i$

Notăm: $e^{iz} = t$

$\Rightarrow 2t + \frac{2}{t} = 3+i \mid \cdot t \Rightarrow 2t^2 - t(3+i) + 2 = 0$

$\Delta = 8+6i-16-6i-8 = -9+6i+1 = (3i+1)^2 \Rightarrow \sqrt{\Delta} = 3i+1$

$t_{1,2} = \frac{3+i \pm (3i+1)}{4}$

$t_1 = \frac{3+i+3i+1}{4} = \frac{4i+4}{4} = i+1$

$t_2 = \frac{3+i-3i-1}{4} = \frac{2-2i}{4} = \frac{1}{2} - \frac{1}{2} i$

Caz I $t = 1+i \Rightarrow e^{iz} = 1+i \Rightarrow iz = \operatorname{Ln}(1+i) \Rightarrow z = -i \operatorname{Ln}(1+i) \Rightarrow$

$\Rightarrow z = -i \left[\ln |1+i| + i \left[\operatorname{Arg}(1+i) + 2k\pi \right] \right] ; k \in \mathbb{Z}$

$z = -i \left[\frac{1}{2} \ln 2 + i \left(\operatorname{arctg} 1 + 2k\pi \right) \right] ; k \in \mathbb{Z}$

$z = -i \left[\frac{1}{2} \ln 2 + i \left(\frac{\pi}{4} + 2k\pi \right) \right] ; k \in \mathbb{Z}$

$z = \frac{\pi}{4} + 2k\pi - \frac{i}{2} \ln 2.$

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caz II $t = \frac{1}{2} - \frac{1}{2}i \Rightarrow e^{iz} = \frac{1}{2} - \frac{1}{2}i \Rightarrow iz = \ln\left(\frac{1}{2} - \frac{1}{2}i\right) \Rightarrow$

$\Rightarrow z = -i \ln\left(\frac{1}{2} - \frac{1}{2}i\right) = -i\left\{\ln\left|\frac{1}{2} - \frac{1}{2}i\right| + i\left[\arg\left(\frac{1}{2} - \frac{1}{2}i\right) + 2k\pi\right]\right\}; k \in \mathbb{Z}$

$z = -i\left[\ln\frac{1}{\sqrt{2}} + i\left(\operatorname{arctg}(-1) + 2k\pi\right)\right]; k \in \mathbb{Z}$

$z = -i\left[-\frac{1}{2}\ln 2 + i\left(-\frac{\pi}{4} + 2k\pi\right)\right]; k \in \mathbb{Z}$

$z = -\frac{\pi}{4} + 2k\pi + \frac{i}{2}\ln 2$

(f) $\operatorname{ch} z = \frac{1}{2} \Leftrightarrow \frac{e^z + e^{-z}}{2} = \frac{1}{2} \Leftrightarrow e^z + e^{-z} = 1$

Notăm: $e^z = t$

$\Rightarrow t + \frac{1}{t} = 1 \Rightarrow t^2 - t + 1 = 0$

$\Delta = 1 - 4 = -3 \Rightarrow \sqrt{\Delta} = i\sqrt{3}$

$\Rightarrow t_{1,2} = \frac{1 \pm i\sqrt{3}}{2}$

caz I $t = \frac{1}{2} + \frac{\sqrt{3}}{2}i \Rightarrow e^z = \frac{1}{2} + \frac{\sqrt{3}}{2}i \Rightarrow z = \ln\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) \Rightarrow$

$\Rightarrow z = \left\{\ln\left|\frac{1}{2} + \frac{\sqrt{3}}{2}i\right| + i\left(\arg\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + 2k\pi\right) \,\middle|\, k \in \mathbb{Z}\right\} =$

$= \left\{\ln 1 + i\left(\operatorname{arctg}\sqrt{3} + 2k\pi\right) \,\middle|\, k \in \mathbb{Z}\right\} =$

$= \left\{i\left(\frac{\pi}{3} + 2k\pi\right) \,\middle|\, k \in \mathbb{Z}\right\}$

caz II $t = \frac{1}{2} - \frac{\sqrt{3}}{2}i \Rightarrow e^z = \frac{1}{2} - \frac{\sqrt{3}}{2}i \Rightarrow z = \ln\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) \Rightarrow$

$\Rightarrow z = \left\{\ln\left|\frac{1}{2} - \frac{\sqrt{3}}{2}i\right| + i\left[\arg\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) + 2k\pi\right] \,\middle|\, k \in \mathbb{Z}\right\} =$

$= \left\{\ln 1 + i\left(\operatorname{arctg}(-\sqrt{3}) + 2k\pi\right) \,\middle|\, k \in \mathbb{Z}\right\} =$

$= \left\{i\left(-\frac{\pi}{3} + 2k\pi\right) \,\middle|\, k \in \mathbb{Z}\right\}$

(g) $z^2 + 2i = 0 \Rightarrow z^2 = -2i \Rightarrow z = \sqrt{-2i}$

$w \overset{not}{=} -2i \Rightarrow |w| = 2$

$w \in Ox$ (pe partea negativă) $\Rightarrow \arg(w) = -\frac{\pi}{2}$

$\Rightarrow z_k = \sqrt{-2i} = \left\{\sqrt{2}\left[\cos\left(\frac{-\frac{\pi}{2} + 2k\pi}{2}\right) + i\sin\left(\frac{-\frac{\pi}{2} + 2k\pi}{2}\right)\right] \,\middle|\, k \in \{0, 1\}\right\}$

$z_0 = \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right) = \sqrt{2}\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = 1 - i$

$z_1 = \sqrt{2}\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right) = \sqrt{2}\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -1 + i$

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